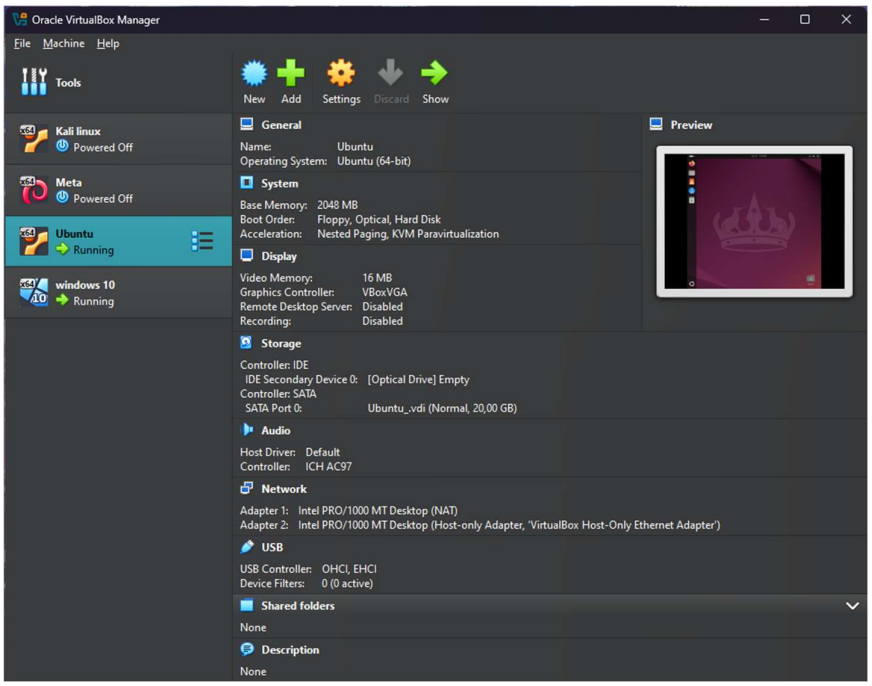
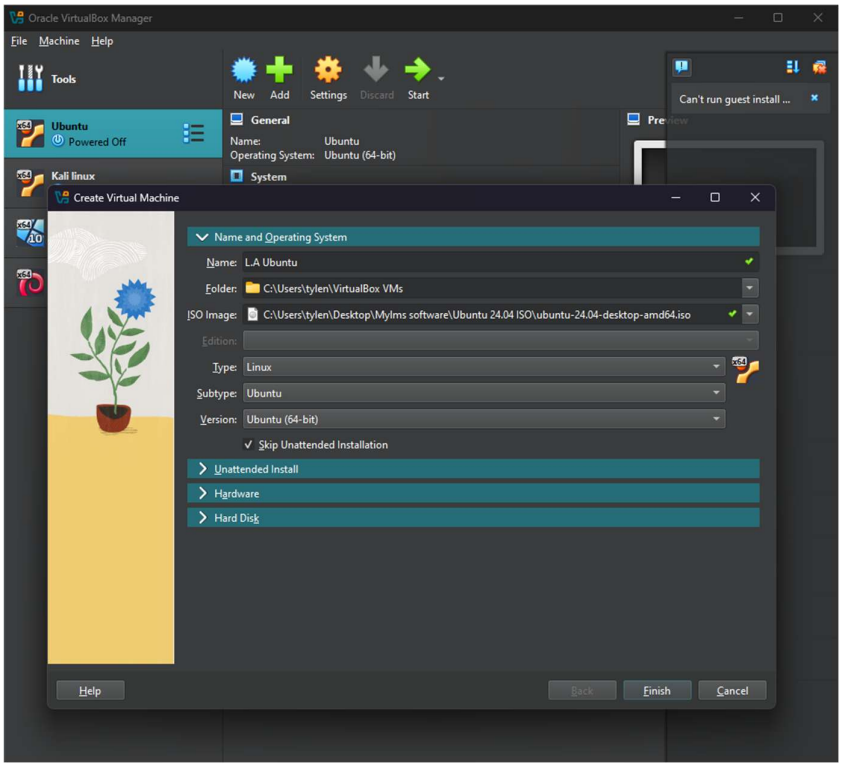
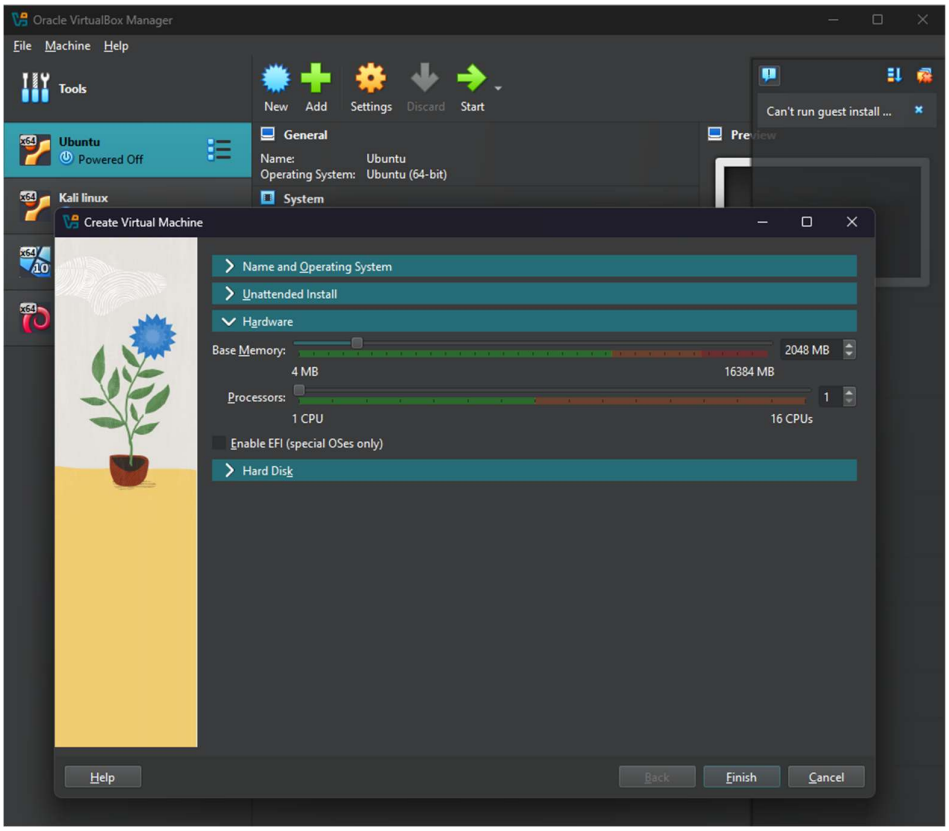
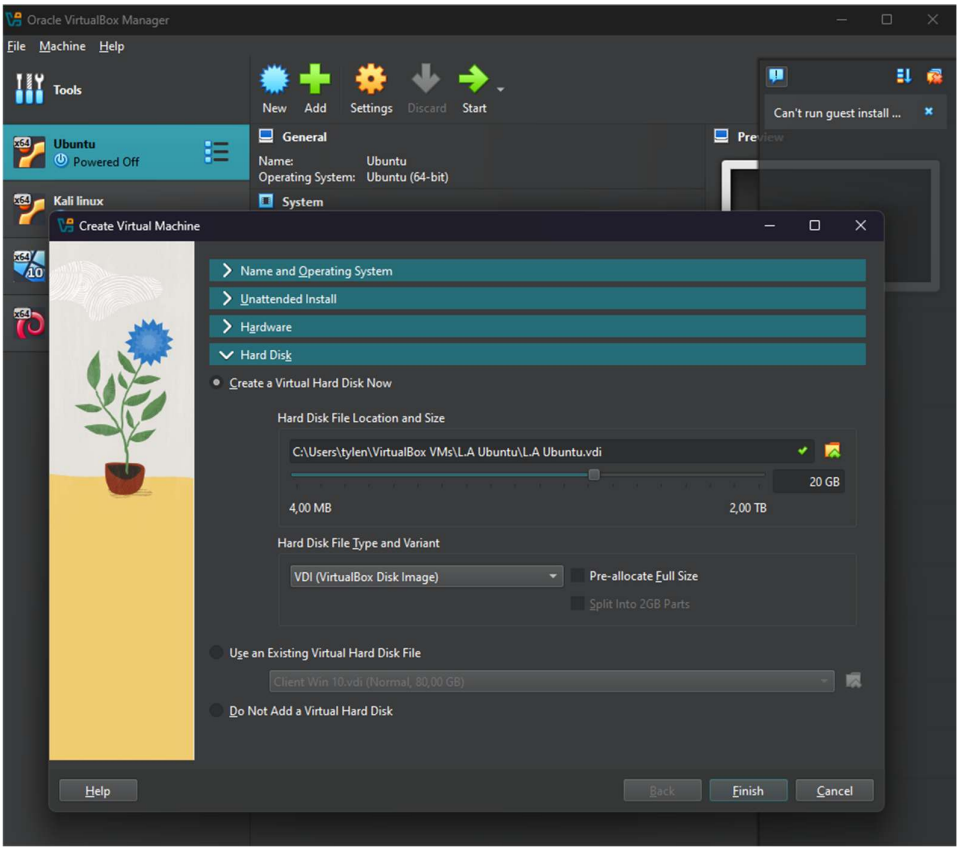
Install VirtualBox



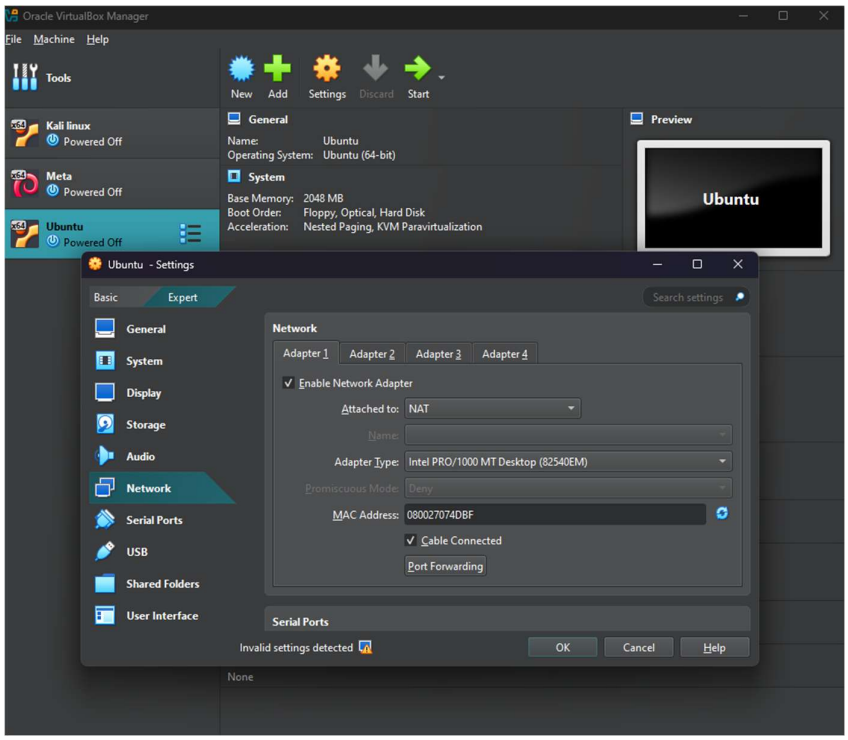
Create and Configure Ubuntu VM



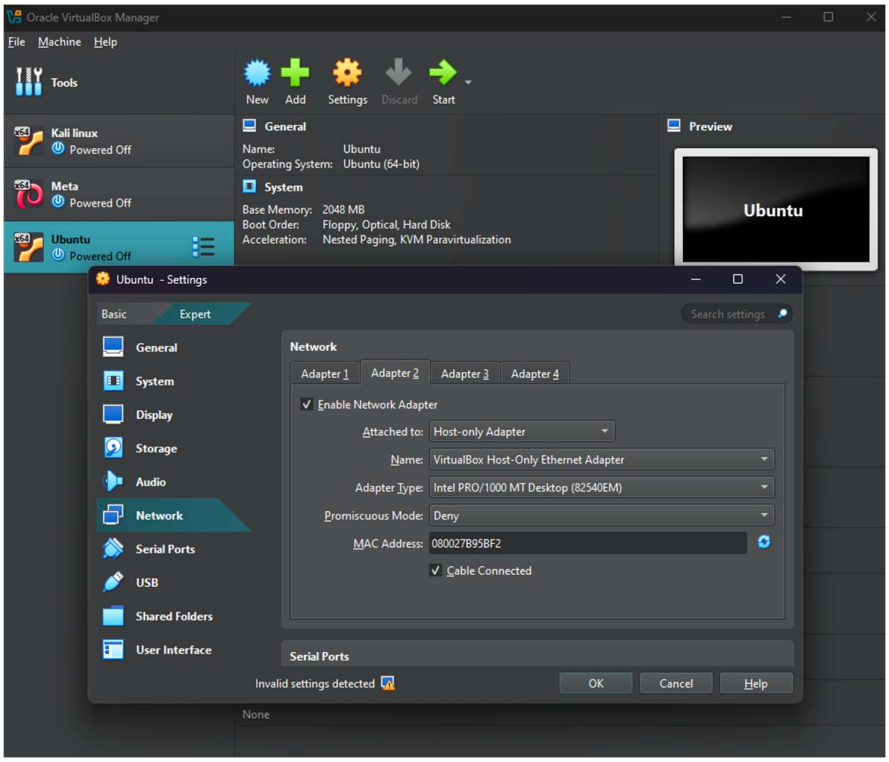

Oracle VirtualBox Manager
File Machine Help
Tools
New Add Settings Discard Start
Can't run guest install ...
Ubuntu
Powered Off
Kali linux
General
Name: Ubuntu
Operating System: Ubuntu (64-bit)
System
Create Virtual Machine
Name and Operating System
Unattended Install
Hardware
Base Memory: 2048 MB
4 MB
16384 MB
Processors: 1
1 CPU
16 CPUs
Enable EFI (special OSes only)
Hard Disk
Help
Back
Finish
Cancel


Oracle VirtualBox Manager
File Machine Help
Tools
New Add Settings Discard Start
Can't run guest install ...
Ubuntu
Powered Off
Kali linux
General
Name: Ubuntu
Operating System: Ubuntu (64-bit)
System
Create Virtual Machine
Name and Operating System
Unattended Install
Hardware
Hard Disk
Create a Virtual Hard Disk Now
Hard Disk File Location and Size
C:\Users\tylen\VirtualBox VMs\L.A Ubuntu\L.A Ubuntu.vdi
20 GB
4,00 MB
2,00 TB
Hard Disk File Type and Variant
VDI (VirtualBox Disk Image)
Pre-allocate Full Size
Split Into 2GB Parts
Use an Existing Virtual Hard Disk File
Client Win 10.vdi (Normal, 80,00 GB)
Do Not Add a Virtual Hard Disk
Help
Back
Finish
Cancel

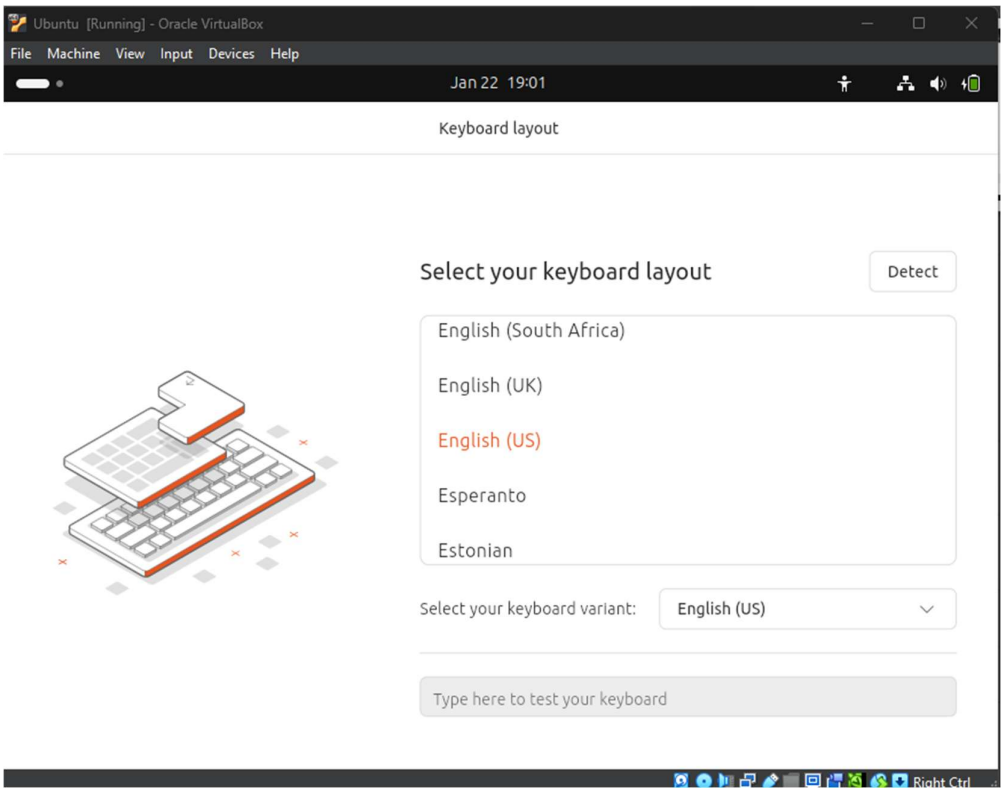
Set Adapter 1 to "NAT" (for internet access).

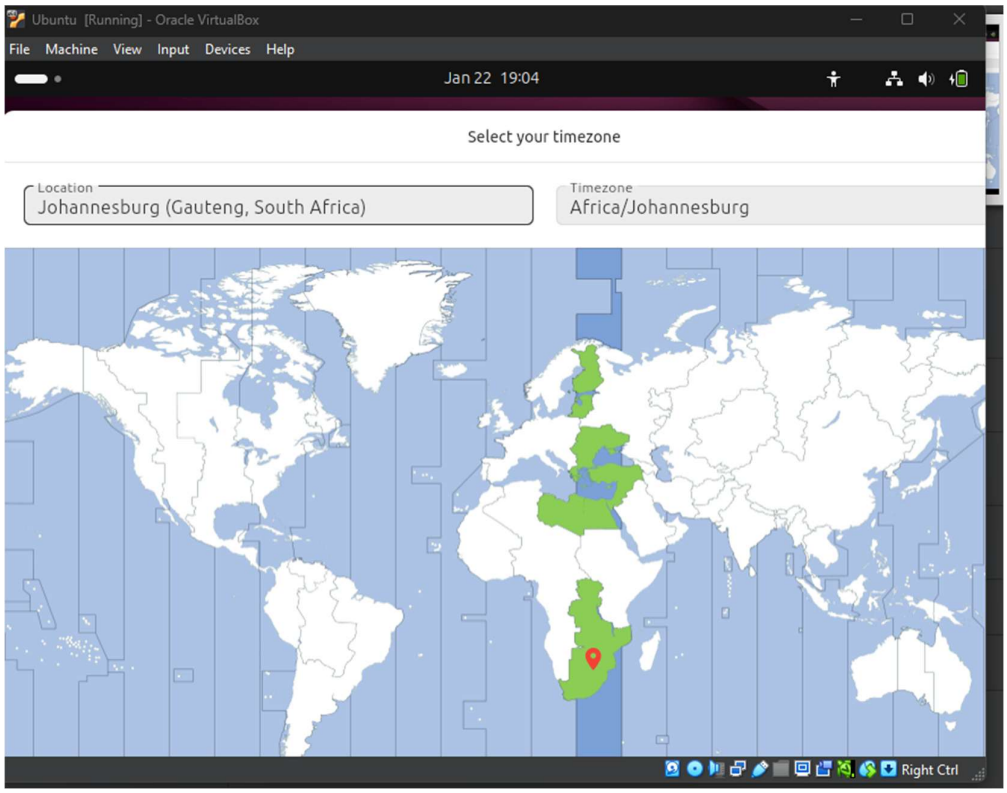


Enable Adapter 2 and set it to Host-only Adapter.

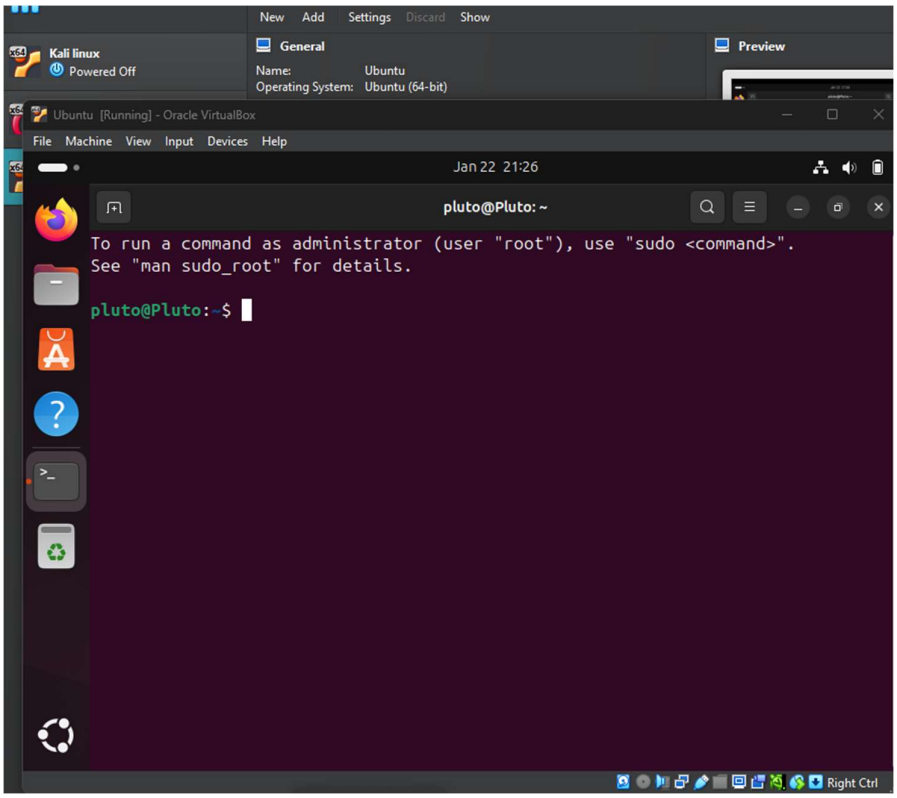


Install Ubuntu OS



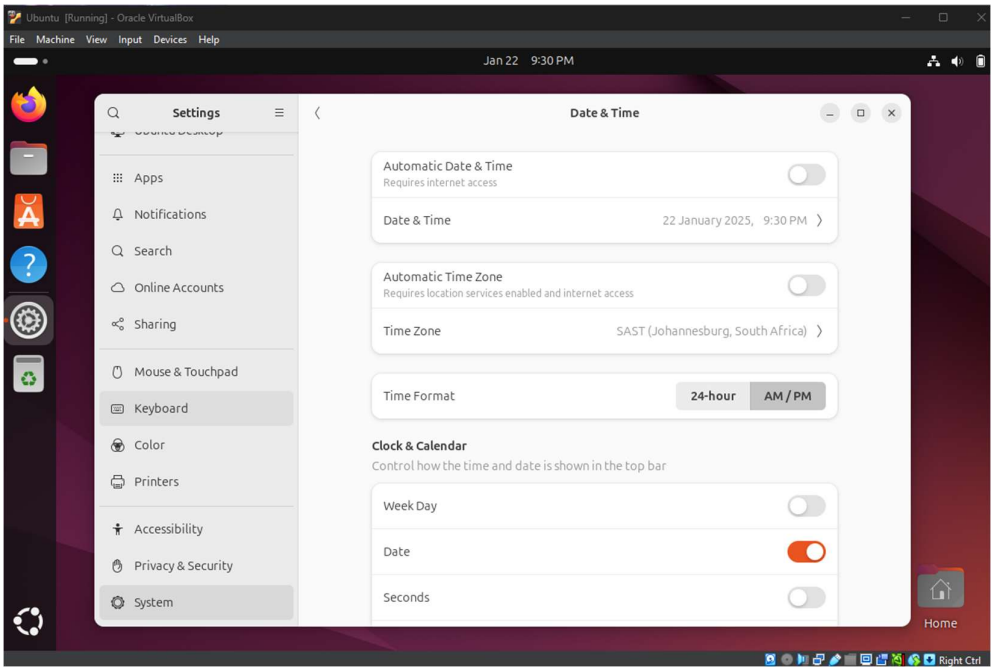


Set the hostname for the Linux Machine to **pluto**.

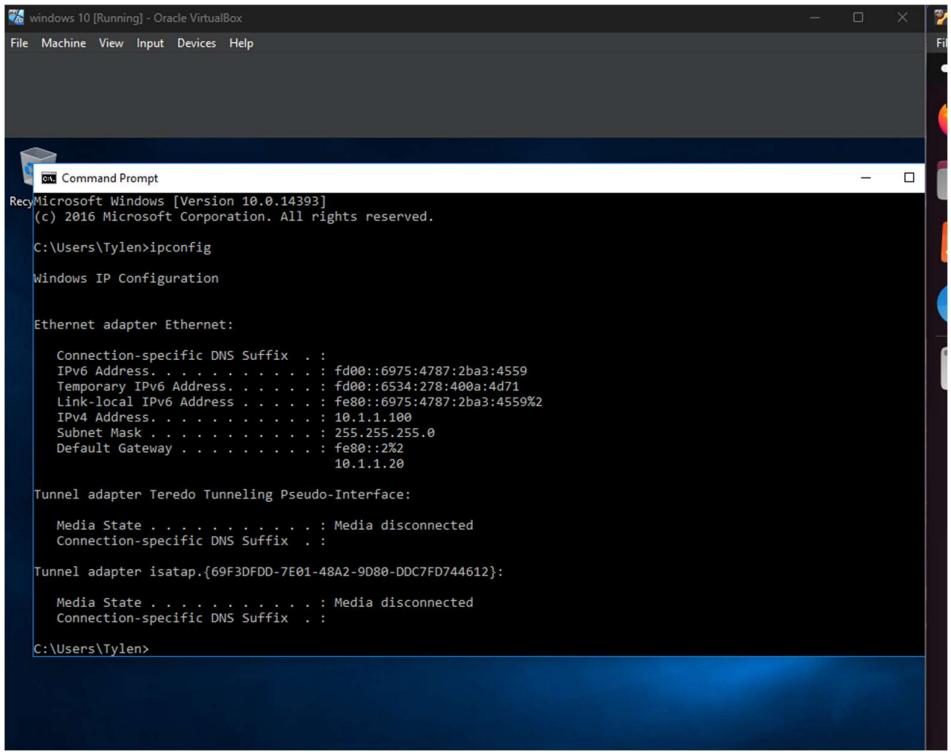


Time to be set manually.

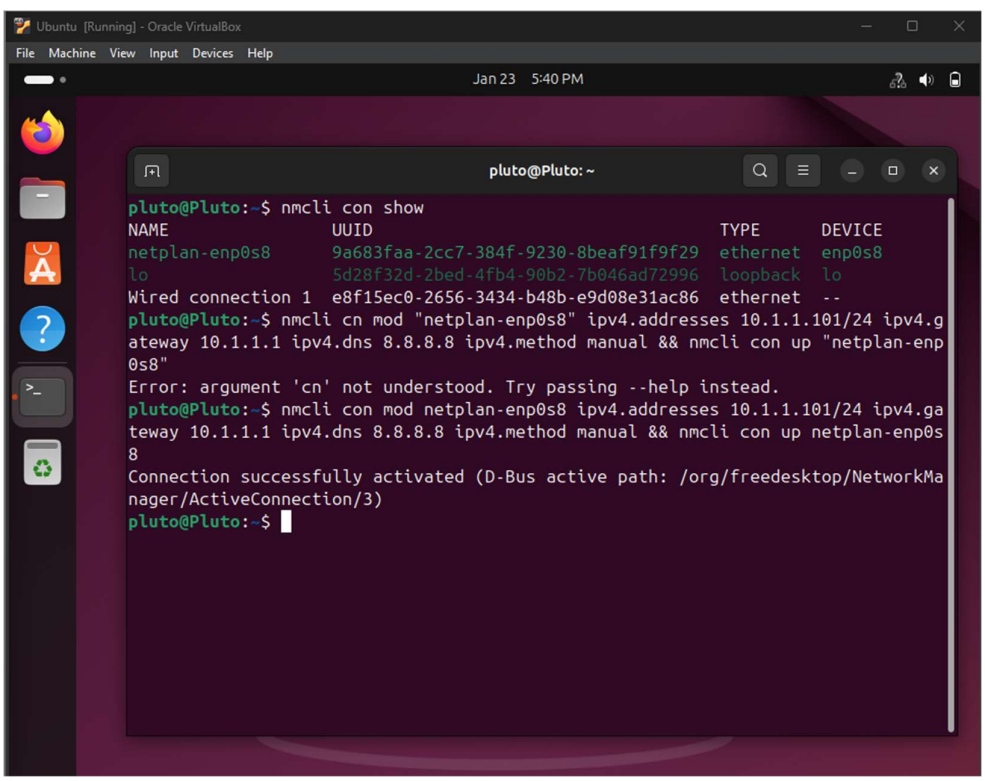
Clock to show in 12-hour time



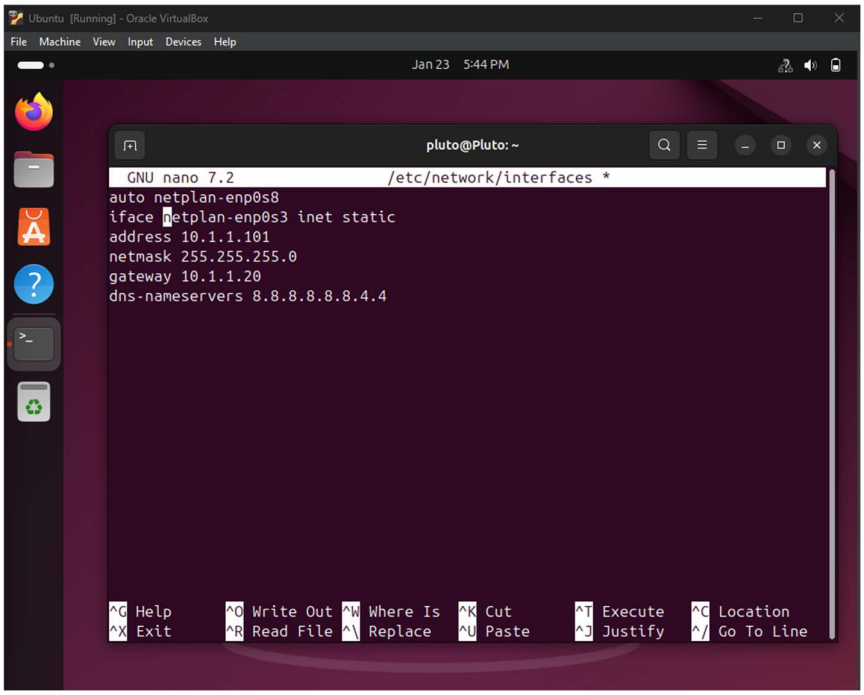
Assign Windows machine (10.1.1.100)



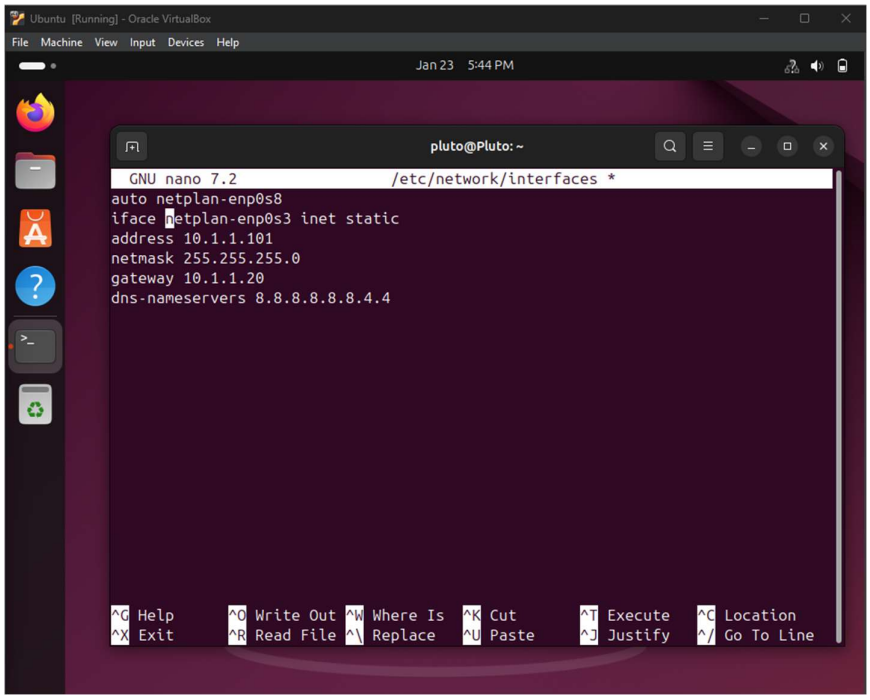
Using a single command, enable the network card and assign a static IP address to the Linux Machine. The address should be in the same range as that of the Windows 10 Machine.



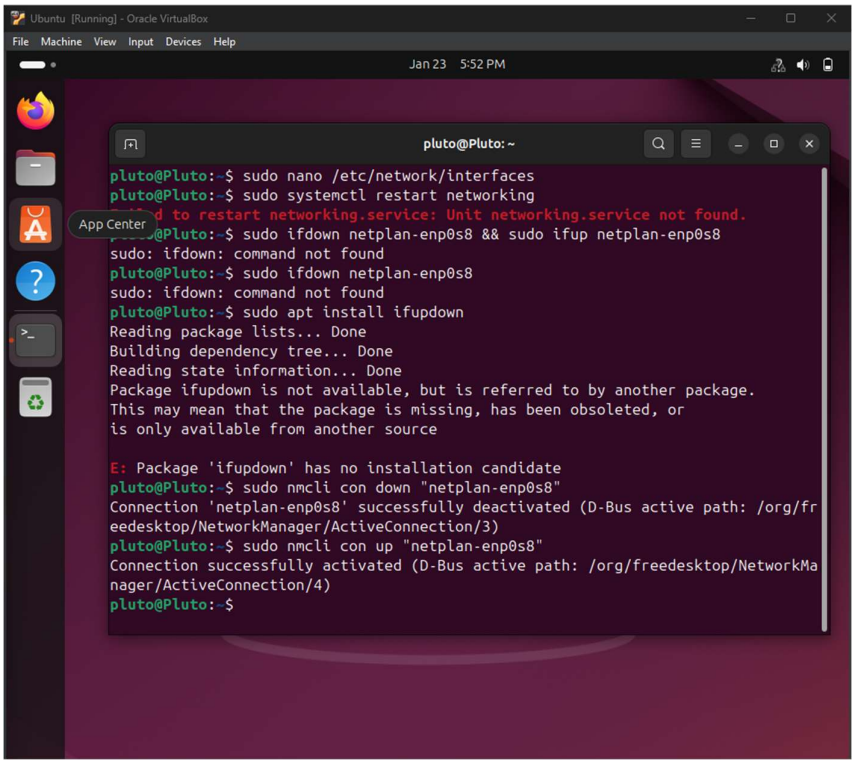
Edit the network interfaces configuration file so that the file reflects the static IP address that has been assigned to the system. Entries should also be added to specify the subnet mask and the gateway.



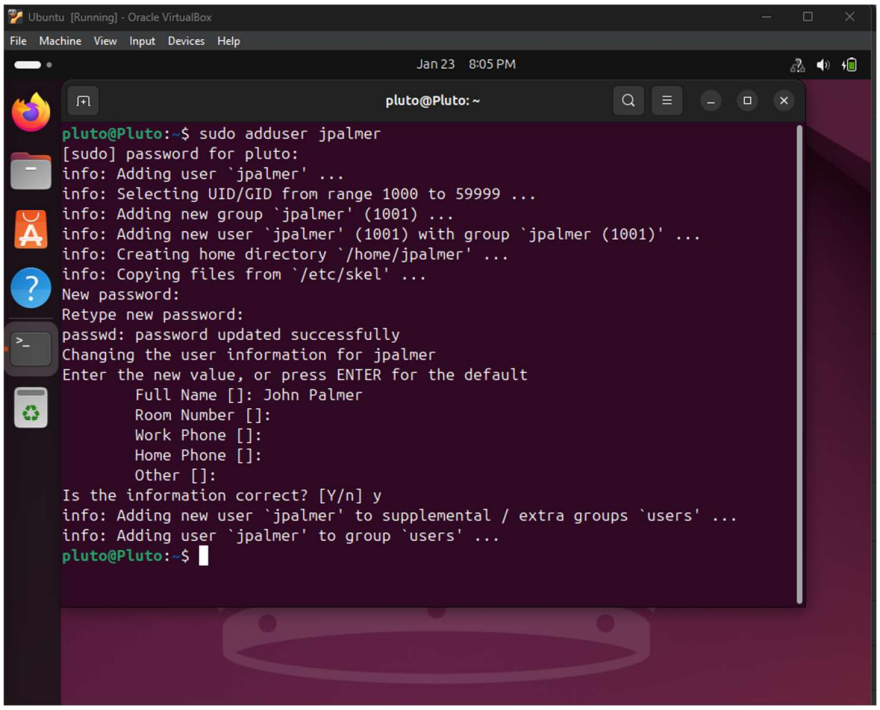
The gateway address must be set to **10.1.1.20**.



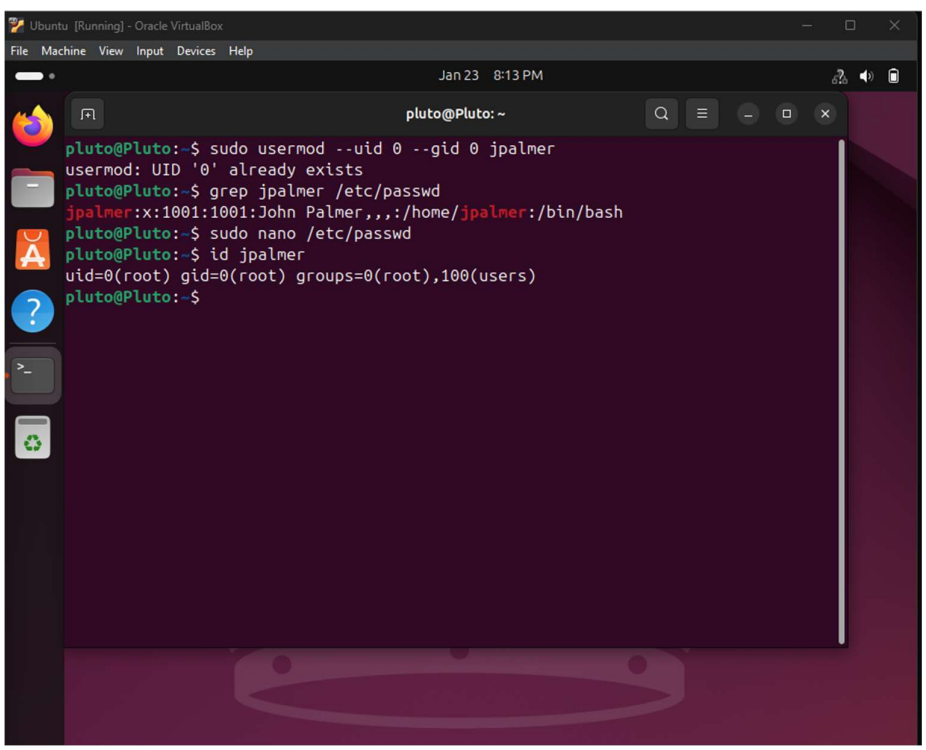
Restart the networking service.



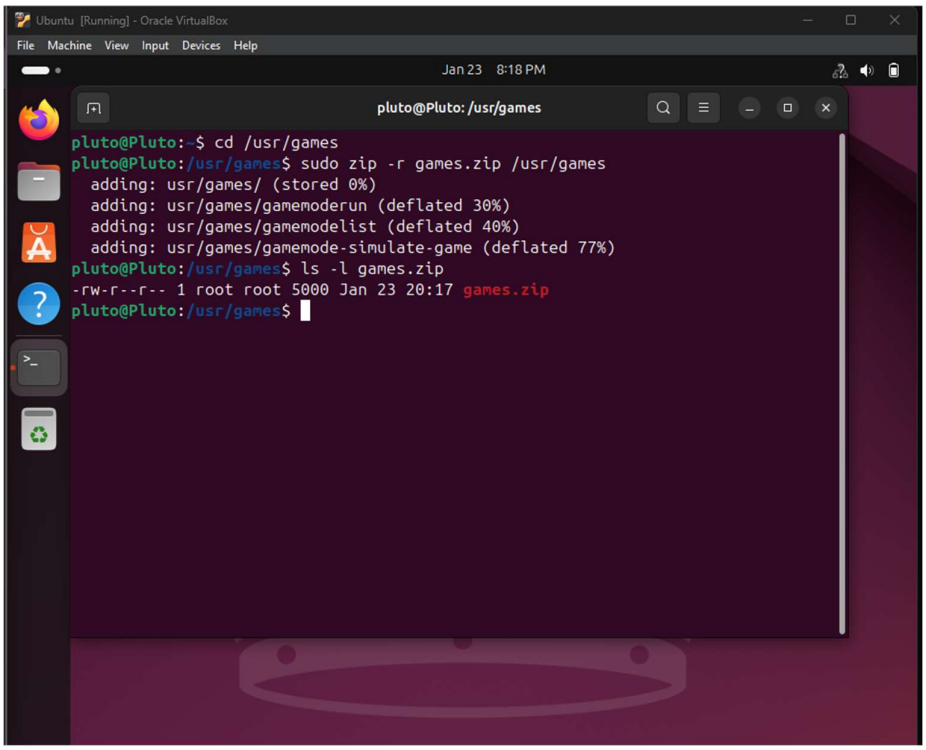
Add an additional user account to the Linux System for John Palmer. The username must be set to jpalmer and a suitable password must be specified.



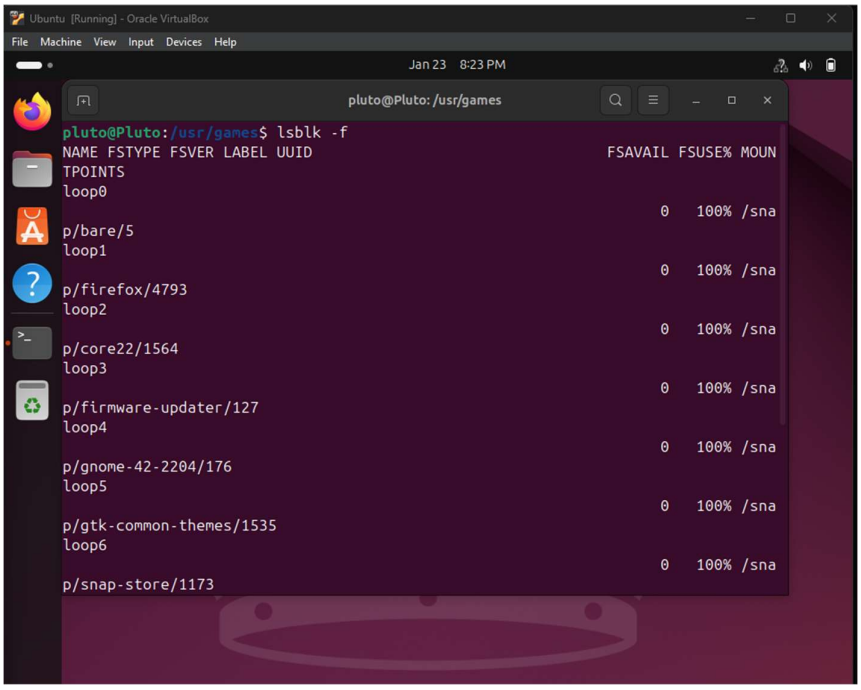
Using a single command, change the **User ID** and **Group ID** of John Palmer to that of the root account.



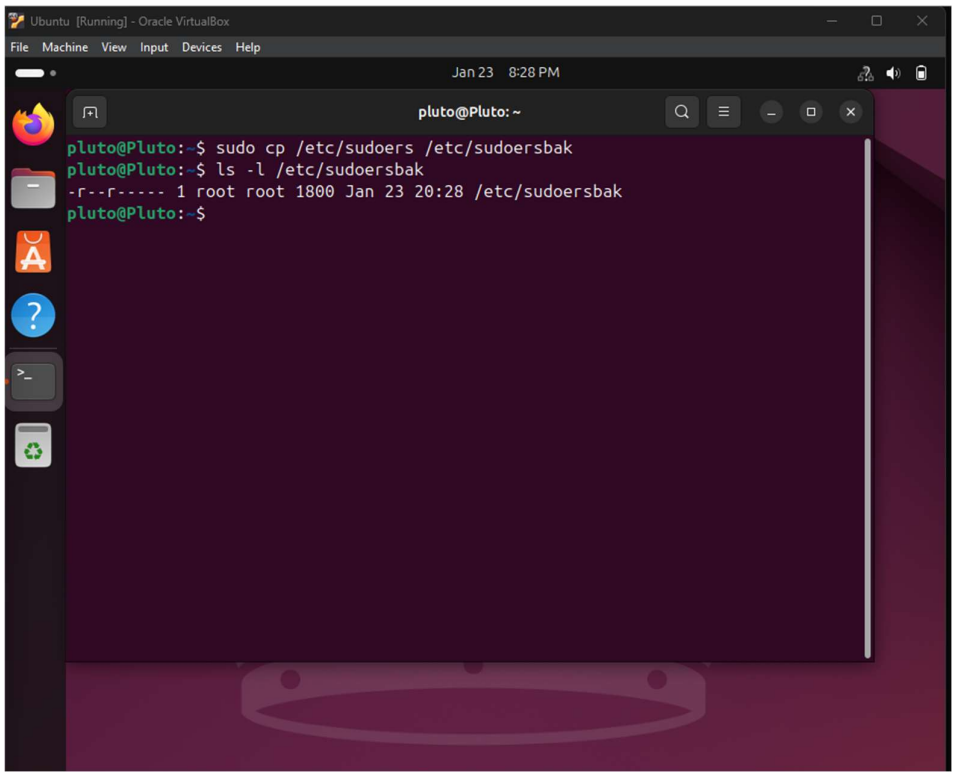
Change to the **/usr/games** directory. Using a single command, compress the contents of the directory into a format that will be compatible with a Windows operating system. The compressed file should be called **games**.



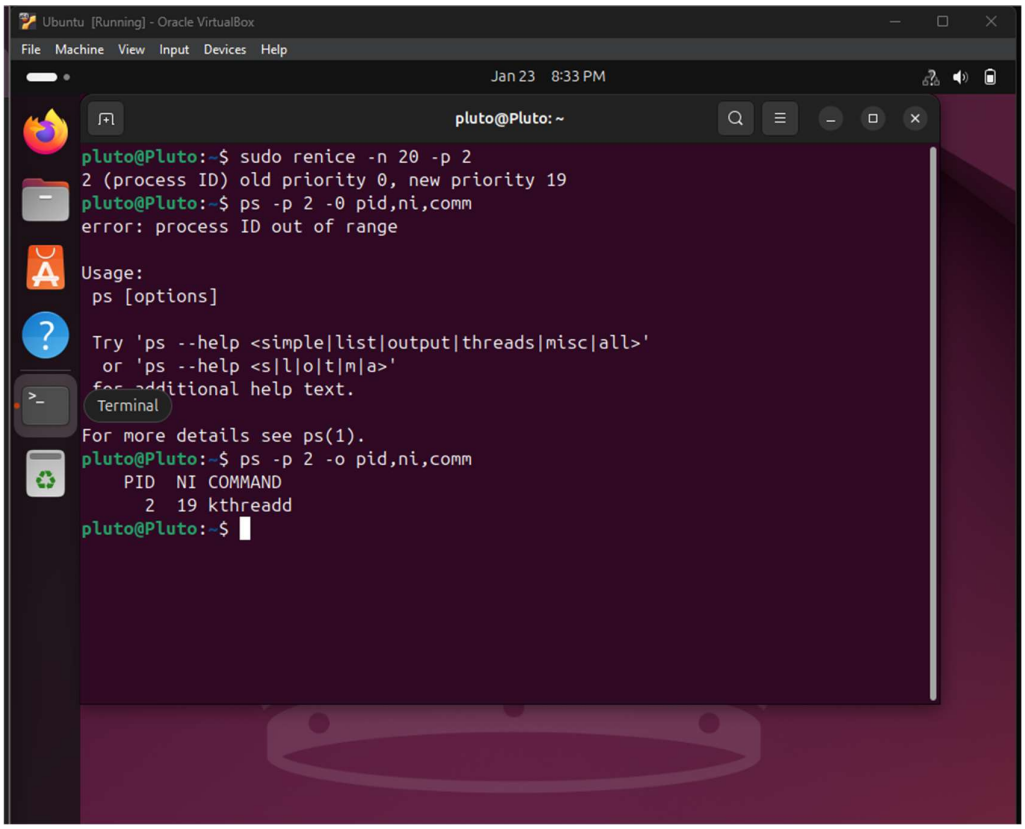
Determine the current operational state of the device, **sda5**. Write the results on a piece of paper.



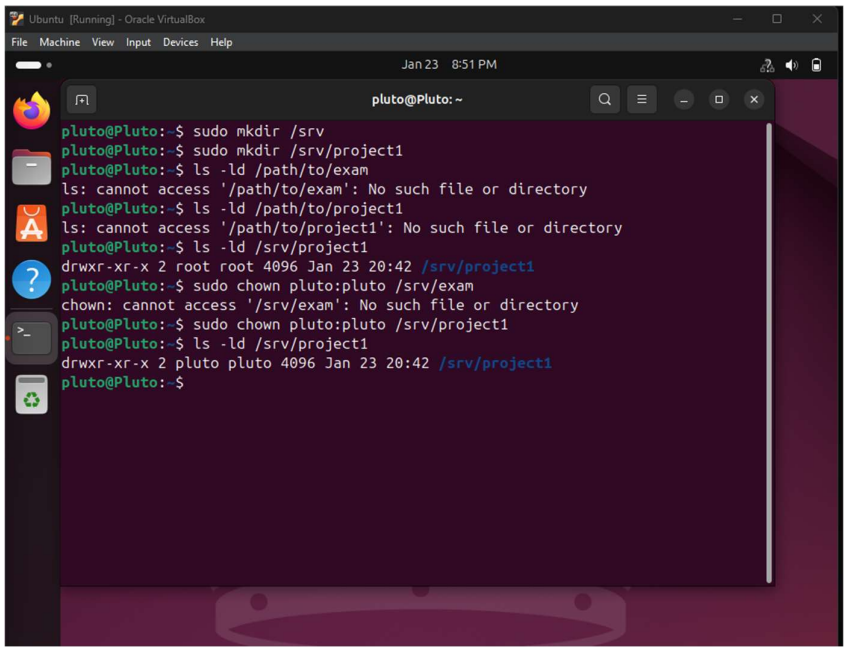
Create a backup copy of the **sudoers** file called **sudoersbak** in the **/etc** directory.



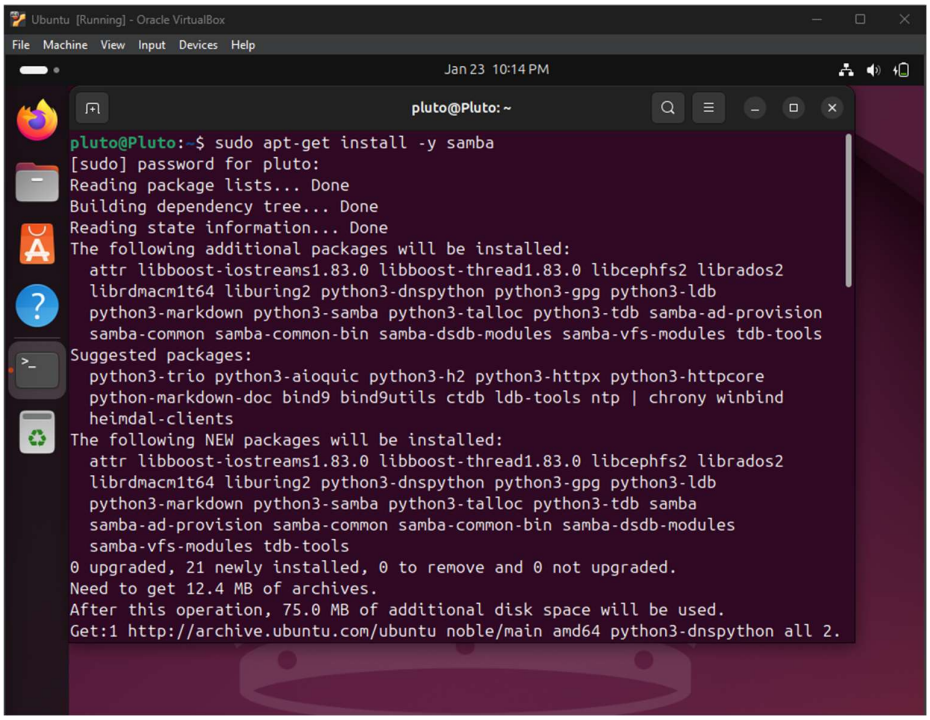
Set the nice number of the **kthreadd** process (**PID 2**) to **20**.

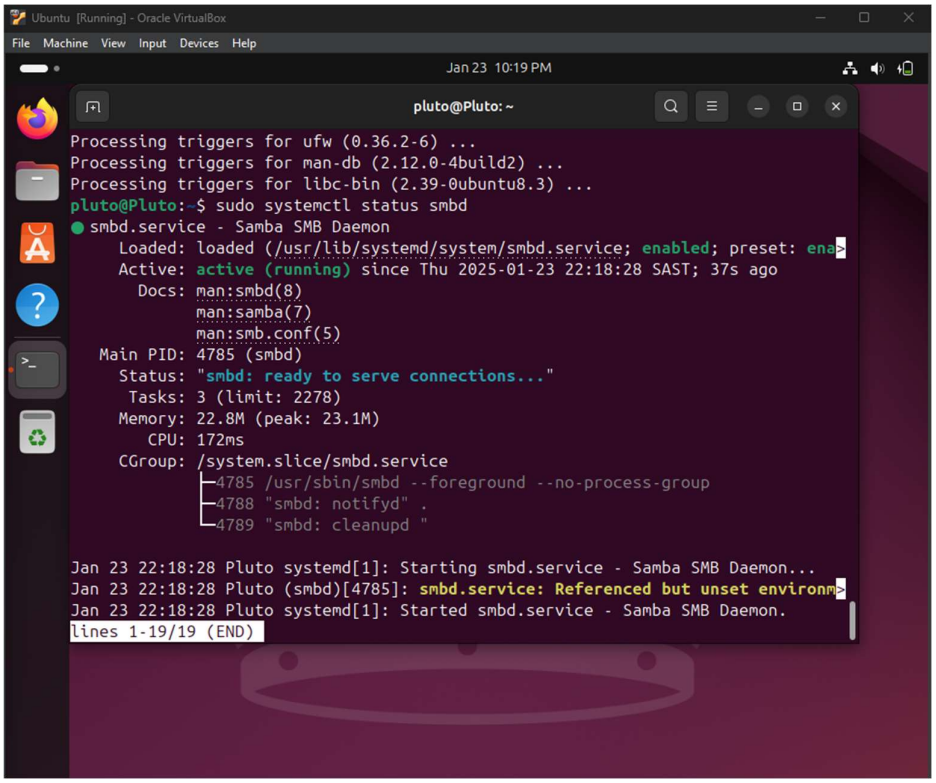


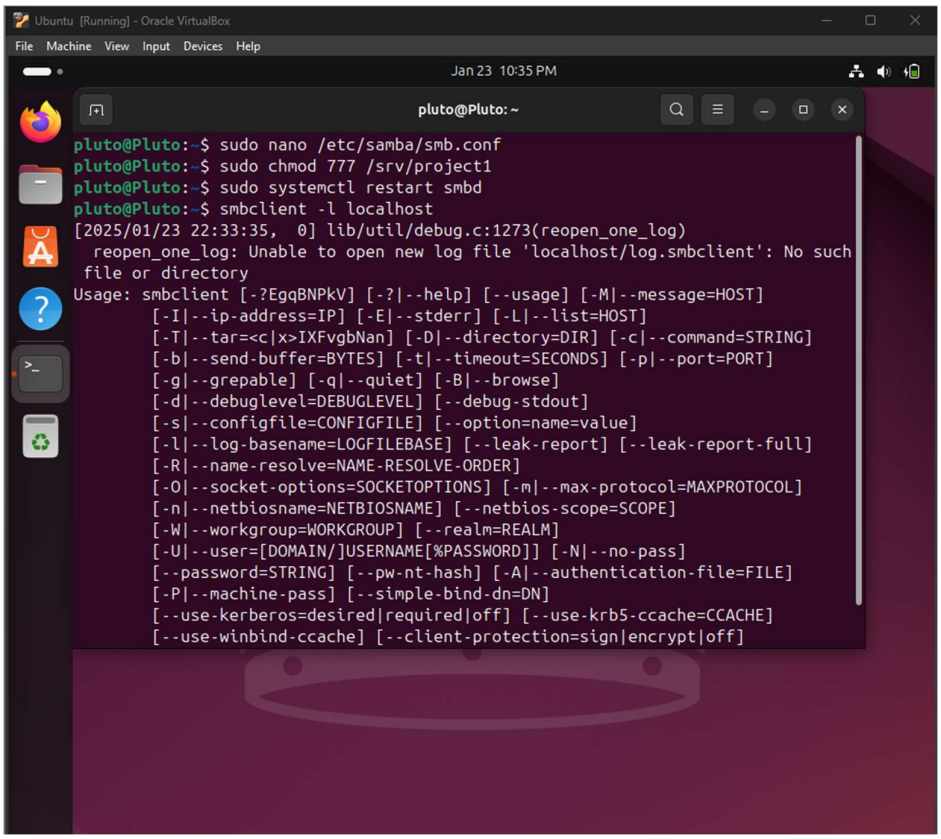
On the Linux Terminal, create a new directory called srv directory. Create a directory called project1 in the /srv directory. Verify that you are the owner of the exam directory and that users can write to this directory.



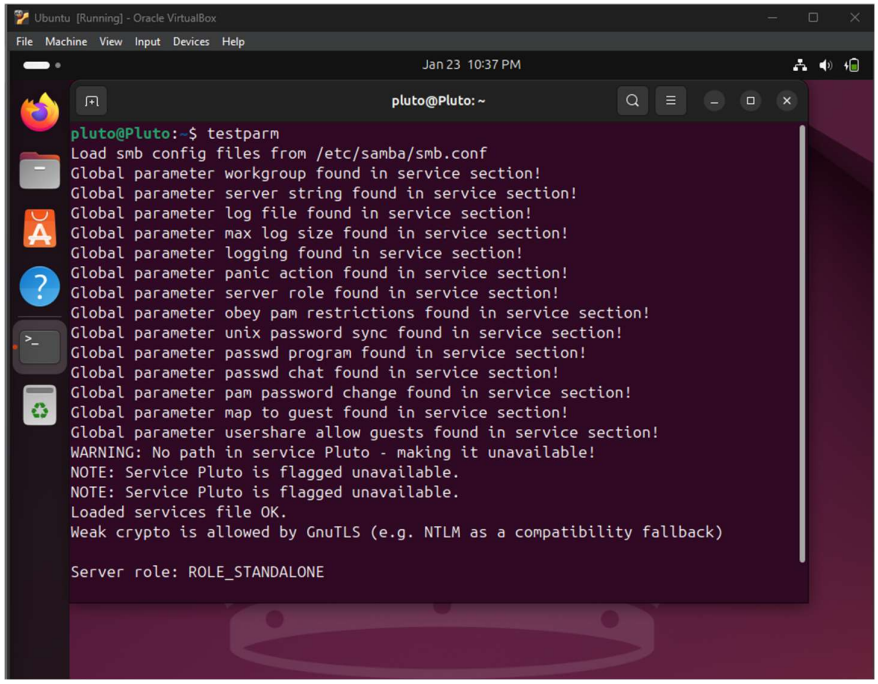
In the Linux Terminal, run the necessary sudo apt-get command then install the necessary sharing services and software. Share the **project1** directory on the Linux System and set the name of the share to your name.



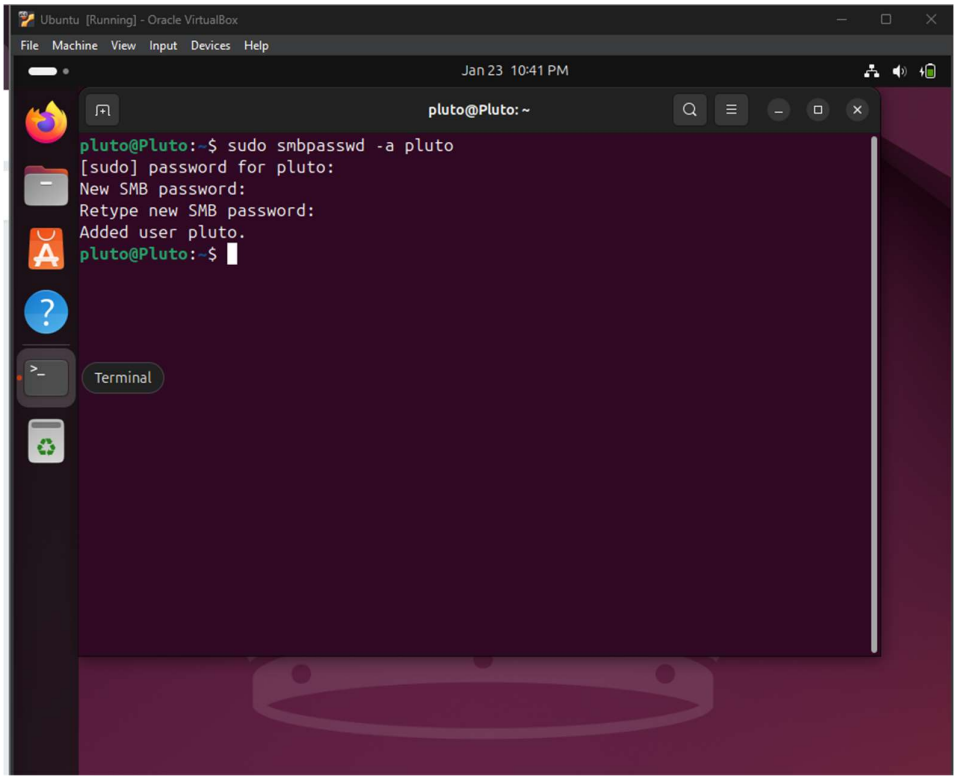

Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 23 10:19 PM
pluto@Pluto: ~
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
pluto@Pluto:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: ena
Active: active (running) since Thu 2025-01-23 22:18:28 SAST; 37s ago
Docs: man:smbd(8)
man:samba(7)
man:smb.conf(5)
Main PID: 4785 (smbd)
Status: "smbd: ready to serve connections..."
Tasks: 3 (limit: 2278)
Memory: 22.8M (peak: 23.1M)
CPU: 172ms
CGroup: /system.slice/smbd.service
├─4785 /usr/sbin/smbd --foreground --no-process-group
├─4788 "smbd: notifyd" .
└─4789 "smbd: cleanupd "
Jan 23 22:18:28 Pluto systemd[1]: Starting smbd.service - Samba SMB Daemon...
Jan 23 22:18:28 Pluto (smbd)[4785]: smbd.service: Referenced but unset environm
Jan 23 22:18:28 Pluto systemd[1]: Started smbd.service - Samba SMB Daemon.
lines 1-19/19 (END)


Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 23 10:35 PM
pluto@Pluto: ~
pluto@Pluto:~$ sudo nano /etc/samba/smb.conf
pluto@Pluto:~$ sudo chmod 777 /srv/project1
pluto@Pluto:~$ sudo systemctl restart smbd
pluto@Pluto:~$ smbclient -l localhost
[2025/01/23 22:33:35, 0] lib/util/debug.c:1273(reopen_one_log)
reopen_one_log: Unable to open new log file 'localhost/log.smbclient': No such file or directory
Usage: smbclient [-?EgqBNPkV] [-?|--help] [--usage] [-M|--message=HOST]
[-I|--ip-address=IP] [-E|--stderr] [-L|--list=HOST]
[-T|--tar=<c|x>IXFvgbNan] [-D|--directory=DIR] [-c|--command=STRING]
[-b|--send-buffer=BYTES] [-t|--timeout=SECONDS] [-p|--port=PORT]
[-g|--grepable] [-q|--quiet] [-B|--browse]
[-d|--debuglevel=DEBUGLEVEL] [--debug-stdout]
[-s|--configfile=CONFIGFILE] [--option=name=value]
[-l|--log-basename=LOGFILEBASE] [--leak-report] [--leak-report-full]
[-R|--name-resolve=NAME-RESOLVE-ORDER]
[-O|--socket-options=SOCKETOPTIONS] [-m|--max-protocol=MAXPROTOCOL]
[-n|--netbiosname=NETBIOSNAME] [--netbios-scope=SCOPE]
[-W|--workgroup=WORKGROUP] [--realm=REALM]
[-U|--user=[DOMAIN/]USERNAME[%PASSWORD]] [-N|--no-pass]
[--password=STRING] [--pw-nt-hash] [-A|--authentication-file=FILE]
[-P|--machine-pass] [--simple-bind-dn=DN]
[--use-kerberos=desired|required|off] [--use-krb5-ccache=CCACHE]
[--use-winbind-ccache] [--client-protection=sign|encrypt|off]

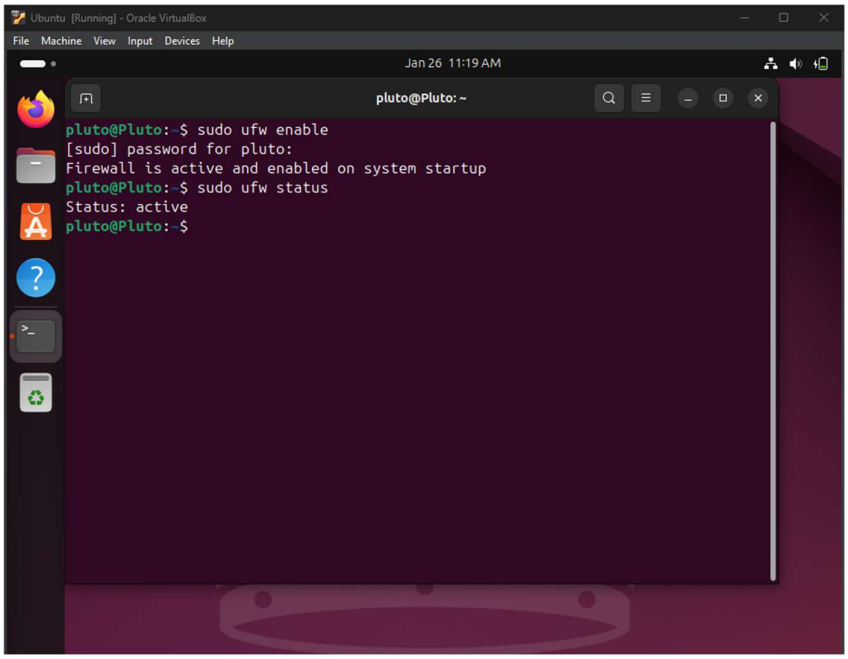
Test the smb.conf file for any syntax errors and restart the Samba Service.



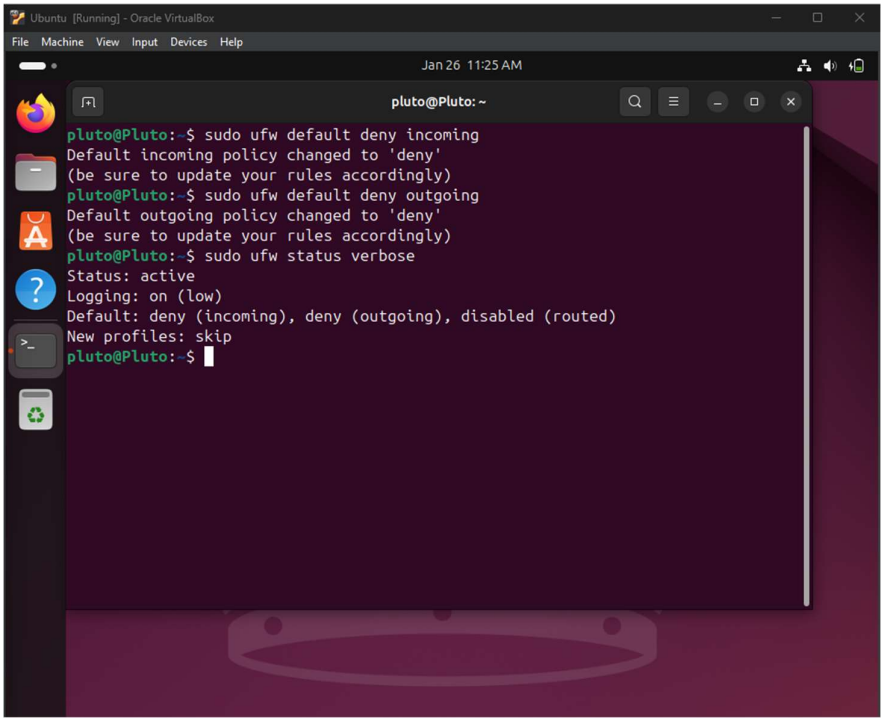
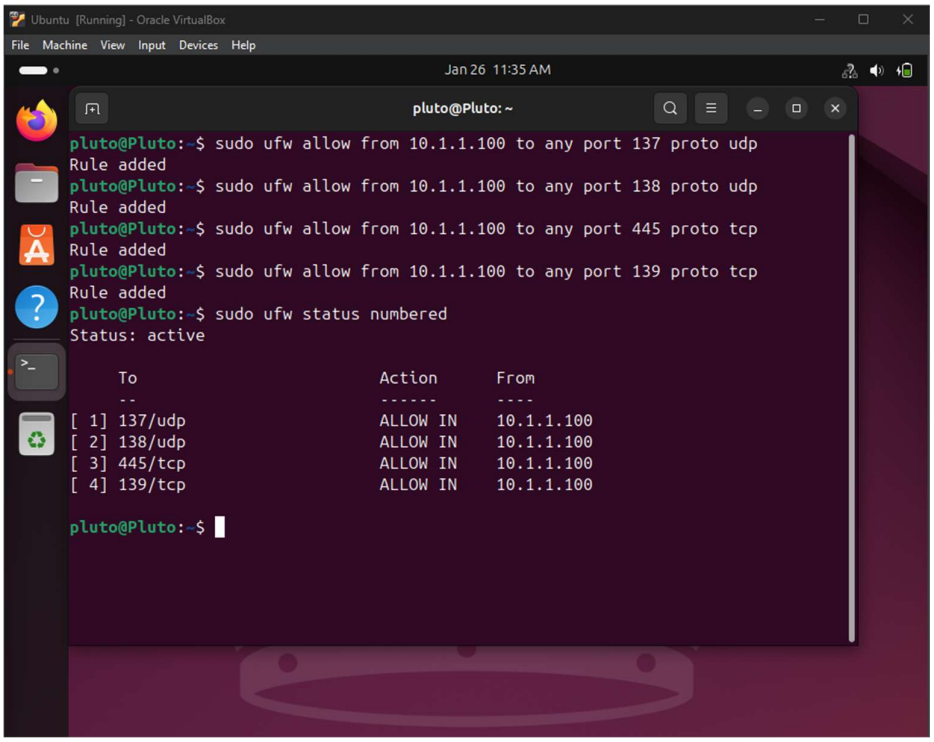
Add a Samba user account for yourself on the Linux system with the password set to password246.



Enable the ufw firewall and confirm the current status of ufw.



Set the ufw default policy to deny all traffic.



Add four rules to allow traffic to pass through the firewall via the four Samba ports. The rules must allow traffic from only the Windows 10 machine to pass through.